

RAIDA FARDOUS

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PROFESSIONAL SUMMARY

Computer Engineering student at the University of Toronto with 16 months of experience in SoC design integration and verification at AMD. Experienced in formal verification, RTL debugging, embedded systems, and automation using Python and C/C++. Interested in ASIC Design, Design Verification, and Hardware Engineering roles.

EDUCATION

University of Toronto

September 2021 - April 2027

Bachelor's, Computer Engineering

Will obtain Bachelor's of Applied Science and Engineering in April 2027.

SKILLS

Programming: Python, C/C++, C#, HTML/CSS, ARM Assembly, Verilog, VHDL, Perl, Ruby, Linux/Unix

Tools: Synopsys Verdi, Synopsys Formality, DVLEC, Git, JIRA, Perforce, Simulink, MATLAB, LTSpice, Altium

Skills: Design Verification, Embedded Systems, Hardware Design, ASIC Design, DV, RTL Verification

PROFESSIONAL EXPERIENCE

Advanced Micro Devices (AMD)

May 2024 - August 2025

Silicon Design Integration & Verification Engineer Intern

Markham, Canada

- Design equivalency checks between **SoC design and ASIC IP design** and debug blackbox issues using **formal verification** and **design verification** tools (ie, Verdi, Formal, DVLEC)
- Investigated RTL and integration issues using **Verilog/SystemVerilog**, design documentation, and waveform-based debugging tools
- Integrated and validated firmware for a microprocessor-based I/O subsystem (CPU), supporting firmware-hardware interactions and DMA-based communication within a large-scale SoC design
- Conducted connectivity and integration checks for level-BIA interfaces across multiple **SoC IP blocks**
- Developed **Python, C/C++, Perl, and Bash/Shell automation scripts** to improve RTL verification and design integration workflows
- Used JIRA within an agile development environment to coordinate and resolve integration issues across globally distributed teams

Success Tutorial School

May 2023 - Present

Tutor

Toronto, Canada

- Teaching high school level mathematics, calculus & vectors, physics, chemistry, and more
- Adapted directly from the Ontario education curriculum
- Strengthened communication and technical explanation skills by teaching mathematical and scientific concepts to students with diverse learning needs

PROJECTS EXPERIENCE

Microcontroller Radio Design - [Link to project](#)

January 2023 - April 2023

- Skills: IC Design, C Programming, Hardware Design
- Prototyped and assembled a microcontroller and local oscillator as a subsystem of a FLRTRX radio
- Usage of schematic capture through computer aided design (CAD) tools (Altium), electronic circuit simulation, and instrument control to create a printed-circuit board (PCB)
- Designed digital hardware and firmware interfaces, integrating peripheral communication and hardware control logic

Battleship - ARM Assembly Project - [Link to project](#)

January 2023 - April 2023

- Skills: FPGA, C programming, ARM Assembly programming, I/O peripheral and memory mapped addressing, computer architecture, embedded systems
- Program of popular video game using C and ARM Assembly
- Usage of PS/2 keyboard for IO and use of VGA pixel buffer, ARM A9 private timer, and interrupts to implement game logic and visuals

RhythmMaster - Polyrhythm Assistance - [Link to project](#)

February 2023 - March 2023

- Skills: C programming, I/O device integration, Arduino
- Creation of polyrhythm sensor coded in Arduino C/C++ structure
- Physical design creation was with an Arduino Duo microcontroller, force sensor, HEX input, and a liquid crystal display (LCD)
- Program outputs multiple, contrasting drum beats simultaneously to help practice polyrhythm techniques